

Philip Mascia

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EDUCATION

Boston University

Master of Engineering, Biomedical Engineering

Relevant Coursework: BME Programming, Clinical Needs Finding, Biomaterials, Continuum Mechanics

Boston, MA

May 2027

University of Virginia

Bachelor of Science, Biomedical Engineering, GPA: 3.81

Honors: Dean's List all semesters, Student Athlete, Peter Casey Gillcrist Memorial Scholarship - \$10,000

Charlottesville, VA

May 2026

PROJECTS

Cardiac Surgical Efficiency Device | Biomedical Clinical Needs Finding

Sep 2026 – Present

- Serving as Clinical Liaison for a cardiac innovation team, coordinating weekly observations of surgical procedures
- Applying clinical needs-finding methodology to identify inefficiencies in workflow and instrumentation with the goal of translating findings into device conceptualization and design

Human Wearable Moulage for Deep Wound Care | Capstone Project

Aug 2025 – May 2026

- Developed wearable hemorrhage-control device for medic training in wound packing
- Designed silicone wound cavity using CAD and Bambu software, incorporating anatomically accurate geometry
- Programmed Arduino Uno-based system integrating four FSR402 sensors, a peristaltic pump, and LCD display
- Calibrated force sensors ($R^2 > 0.7$) and pump output to 350 mL/min, implementing pressure-based control logic to regulate simulated blood flow

Personal Fitness Wearables: Wellness or Medical Device? | Thesis Research

Aug 2025 – May 2026

- Conducted literature analysis of FDA classification boundaries between wellness wearables and medical devices
- Applied Market Pull Theory and Social Construction of Technology to analyze wearable innovation, regulatory interpretation, and WHOOP blood-pressure insight case

Atrial Fibrillation Detection Model | Biomedical Systems Analysis and Design Final Project

Sep – Dec 2024

- Analyzed R-R and P-P intervals of 12-lead ECG plots to identify presence of atrial fibrillation in patients
- Presented MATLAB code and findings in a conference paper at a symposium

EXPERIENCE

Bioprocess and Biomanufacturing Intern | AgroSpheres

Charlottesville, VA | Jun – Aug 2025

- Operated 1-10L bioreactors for 48-hour bacterial fermentation, optimizing nutrient and chemical concentrations and monitoring cell growth via spectroscopy and serial dilutions
- Harvested and isolated specific slow-release cells from fermentation broth using centrifugation; quantified batch results to assess strain and process performance

Student Observer | UVA Health Prosthetics and Orthotics

Charlottesville, VA | Jan – May 2025

- Observed clinical prosthetist meeting with patients and manufacturing upper and lower limb prosthetic wearables

Undergraduate Research Assistant | Multiscale Muscle Mechanophysiology ("M3") Lab

Sep 2024 – Jan 2025

- Conducted image analysis research mapping DTI scans of striated muscle to advance muscular dystrophy treatments

LEADERSHIP

e-NABLE | Executive Member

Sep 2024 – May 2026

- Designed, printed, and assembled low-cost prosthetic upper limb devices using CAD and Bambu software
- Served as Assembly Lead, spearheading assembly logic of designed prosthetic parts

Cheerleader | UVA Cheerleading

Aug 2023 – May 2026

- Committed 15-30 hours per week of practice and travel for UVA football, basketball, wrestling, and volleyball events
- Awarded Rookie of the Year, Hardest Working, 4.0 Club (2x)

Other Activities: St. Jude Memphis Marathon Gold Hero, Eagle Scout, Sigma Chi Executive Member

TECHNICAL SKILLS

Software: Python, R, Arduino IDE / C++, MATLAB, Spyder, NetLogo, Image Analysis, CAD, Microsoft Office

Laboratory: Cell culturing and growth kinetics, Western Blotting, Microscopy, PCR Amplification, Histology, Spectroscopy